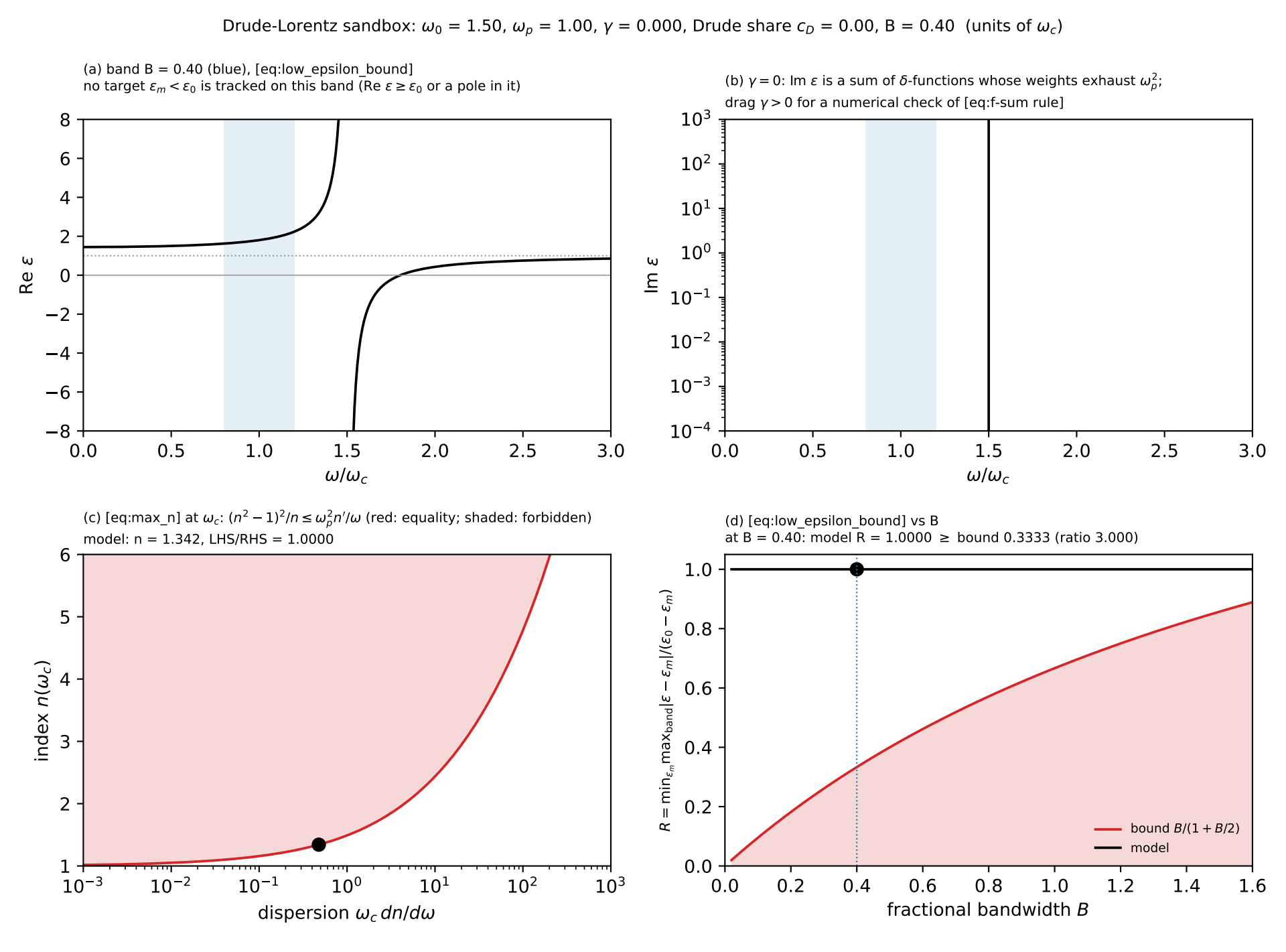
Drude-Lorentz sandbox: ω_0 = 1.50, ω_p = 1.00, γ = 0.000, Drude share c_D = 0.00, B = 0.40 (units of ω_c)
(a) band B = 0.40 (blue), [eq:low_epsilon_bound]
no target $\varepsilon_m < \varepsilon_0$ is tracked on this band (Re $\varepsilon \geq \varepsilon_0$ or a pole in it)
Re ε
ω/ω_c
(b) $\gamma = 0$: Im ε is a sum of δ-functions whose weights exhaust ω_p^2;
drag $\gamma > 0$ for a numerical check of [eq:f-sum rule]
Im ε
ω/ω_c
(c) [eq:max_n] at ω_c: $(n^2-1)^2/n \leq \omega_p^2 n'/\omega$ (red: equality; shaded: forbidden)
model: n = 1.342, LHS/RHS = 1.0000
index $n(\omega_c)$
dispersion $\omega_c\, dn/d\omega$
(d) [eq:low_epsilon_bound] vs B
at B = 0.40: model R = 1.0000 ≥ bound 0.3333 (ratio 3.000)
$R = \min_{\varepsilon_m} \max_{\text{band}} |\varepsilon - \varepsilon_m|/(\varepsilon_0 - \varepsilon_m)$
fractional bandwidth B
bound $B/(1+B/2)$
model